RAJALAKSHMI ENGINEERING COLLEGE

Started on	Friday, 25 September 2026, 8:34 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:19 AM
Time taken	45 mins 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int longestNonDecreasingSubsequence(int arr[], int n) {
    if (n == 0) return 0;
   int *dp = (int *)malloc(n * sizeof(int));
    if (dp == NULL) return -1;
    int max_len = 1;
    for (int i = 0; i < n; i++) {
        dp[i] = 1;
    }
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {
            if (arr[j] <= arr[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }
        if (dp[i] > max_len) {
            max_len = dp[i];
        }
    }

    free(dp);
    return max_len;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int result = longestNonDecreasingSubsequence(arr, n);
    printf("%d\n", result);
     free(arr);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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